

ASG TEMPLATE VALIDATION REPORT

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C. | master: 01_Master_Template/ASG_Master_Template.dxf (sha256 89fd69aabb66024...) | ezdxf 1.4.4, LibreOffice 24.2 (2nd reader), Python | 10.10.2026

Summary: PASS 130 | PARTIAL 1 | UNVERIFIED 13 | FAIL 3. AutoCAD was NOT available: nothing below was tested inside AutoCAD. PASS = tested by re-opening the saved file; PARTIAL = tested with a non-AutoCAD renderer only; UNVERIFIED = needs AutoCAD (steps in README); FAIL = test failed or deliverable missing.

Test procedure

1. Inventory CAD software (AutoCAD, accoreconsole, ODA, LibreDWG, QCAD, BricsCAD).
2. Build all files from asg_standard.py (build_all.py).
3. Re-open master with ezdxf recover + audit, strict parse, and LibreOffice Draw as an independent reader.
4. Compare every header variable, layer, linetype, text/dim/mleader/mline style, scale and layout with the definitions.
5. Check page setup fields, borders, title-block insert scale and attributes, viewport layer/lock/scale/position.
6. Check model space empty, no xrefs/underlays/proxies, only title-block blocks, fonts.
7. Idempotency: re-run scale injection and layout creation; rebuild and compare.
8. Re-open each single-size template and the CTB (binary header + all 255 entries).
9. Build a separate temporary QA drawing: dims of known geometry, viewports 1:20 / 1:5, no-plot objects, lineweight matrix, text heights.
10. Check measured dimension values and rendered text precision; render QA layout at 600 dpi and measure line widths; probe no-plot and viewport-edge areas.
11. In AutoCAD (pending): ASG-SETUP, ASG-QA, plot QA layout to PDF, check annotative scaling, then SAVEAS DWG/DWT.

Results

Area	Test	Status	Evidence
Environment	Native AutoCAD / DWG writer available	FAIL	None of AutoCAD, accoreconsole, ODA File Converter, LibreDWG, QCAD, BricsCAD present
Deliverables	ASG_Master_Template.dwt generated natively	FAIL	Not generated - no DWG writer. Not faked by renaming. Create in AutoCAD per README step 4.
Deliverables	ASG_Master_Template.dwg generated natively	FAIL	Not generated - no DWG writer. Not faked by renaming. Create in AutoCAD per README step 4.
File	Master DXF re-opened (recover + audit)	PASS	0 errors, 0 fixes, AC1032 = AutoCAD 2018 DXF
File	Master DXF strict parse	PASS	ezdxf.readfile without exception
File	Master DXF opened by second, independent reader (LibreOffice Draw)	PASS	converted without load error
File	Opens in AutoCAD	UNVERIFIED	AutoCAD not available - README step 2
Units & variables	\$INSUNITS = 4	PASS	read back 4 - Insertion units = millimetres (no unit conversion on block insert)
Units & variables	\$MEASUREMENT = 1	PASS	read back 1 - Metric - acadiso.lin / acadiso.pat used
Units & variables	\$LUNITS = 2	PASS	read back 2 - Linear units = Decimal
Units & variables	\$LUPREC = 0	PASS	read back 0 - Linear display precision 0 dp
Units & variables	\$AUNITS = 0	PASS	read back 0 - Angular units = Decimal Degrees
Units & variables	\$AUPREC = 2	PASS	read back 2 - Angle precision 0.00
Units & variables	\$ANGBASE = 0.0	PASS	read back 0.0 - 0 deg = East
Units & variables	\$ANGDIR = 0	PASS	read back 0 - Counter-clockwise positive

Area	Test	Status	Evidence
Units & variables	\$LIMMIN = (0.0, 0.0)	PASS	read back (0.0, 0.0) - Nominal limits origin
Units & variables	\$LIMMAX = (200000.0, 150000.0)	PASS	read back (200000.0, 150000.0) - Nominal 200 x 150 m - never restrictive
Units & variables	\$LIMCHECK = 0	PASS	read back 0 - Limits not enforced (large facade elevations)
Units & variables	\$LTSCALE = 1.0	PASS	read back 1.0 - Global linetype scale 1 - dash lengths in paper mm
Units & variables	\$PSLTSCALE = 1	PASS	read back 1 - Viewport scale controls linetype scale
Units & variables	\$CELTSCALE = 1.0	PASS	read back 1.0 - Object linetype scale default
Units & variables	\$LWDISPLAY = 1	PASS	read back 1 - Show lineweights on screen
Units & variables	\$CELWEIGHT = -1	PASS	read back -1 - New objects lineweight ByLayer
Units & variables	\$CECOLOR = 256	PASS	read back 256 - New objects colour ByLayer
Units & variables	\$ORTHOMODE = 0	PASS	read back 0 - Ortho off - polar tracking preferred
Units & variables	\$DIMASSOC = 2	PASS	read back 2 - Associative dimensions follow geometry
Units & variables	\$PLINEGEN = 1	PASS	read back 1 - Continuous linetype pattern along polylines
Units & variables	\$FILLMODE = 1	PASS	read back 1 - Fills displayed
Units & variables	\$MIRRTEXT = 0	PASS	read back 0 - Text not mirrored
Units & variables	\$TEXTSIZE = 2.5	PASS	read back 2.5 - Default text height
Units & variables	\$PSTYLEMODE = 1	PASS	read back 1 - Colour-dependent plot styles (CTB)
Units & variables	\$WORLDVIEW = 1	PASS	read back 1 - UCS follows WCS in views
Units & variables	\$UCSORG = (0.0, 0.0, 0.0)	PASS	read back (0.0, 0.0, 0.0) - UCS origin at WCS origin
Units & variables	\$UCSXDIR = (1.0, 0.0, 0.0)	PASS	read back (1.0, 0.0, 0.0) - UCS = World
Units & variables	\$UCSYDIR = (0.0, 1.0, 0.0)	PASS	read back (0.0, 1.0, 0.0) - UCS = World
Units & variables	\$VISRETAIN = 1	PASS	read back 1 - Xref layer overrides retained
Units & variables	\$PROXYGRAPHICS = 1	PASS	read back 1 - Proxy images saved for interoperability
Units & variables	\$CLAYER = ASG-OUTLINE-PRIMARY	PASS	read back ASG-OUTLINE-PRIMARY - Default current layer (plottable geometry)
Units & variables	\$TEXTSTYLE = ASG-TEXT-ANNO	PASS	read back ASG-TEXT-ANNO - Default text style
Units & variables	\$DIMSTYLE = ASG-DIM-ANNO	PASS	read back ASG-DIM-ANNO - Default dimension style
Units & variables	\$CMLSTYLE = ASG-MLINE-2	PASS	read back ASG-MLINE-2 - Default multiline style
Units & variables	Machine/profile variables (OSMODE, AUTOSNAP, PICKBOX ...)	UNVERIFIED	Not drawing-resident; applied only by opt-in ASG-PROFILE in AutoCAD
Linetypes	HIDDEN loaded	PASS	
Linetypes	HIDDEN2 loaded	PASS	
Linetypes	CENTER loaded	PASS	
Linetypes	CENTER2 loaded	PASS	
Linetypes	DASHED loaded	PASS	
Linetypes	DASHED2 loaded	PASS	
Linetypes	DASHDOT loaded	PASS	
Linetypes	DASHDOT2 loaded	PASS	
Linetypes	PHANTOM loaded	PASS	
Linetypes	PHANTOM2 loaded	PASS	
Linetypes	DOT loaded	PASS	
Linetypes	Continuous loaded	PASS	
Layers	34 ASG layers present	PASS	found 34
Layers	No other layers except system 0 / Defpoints	PASS	['0', 'Defpoints']
Layers	Layer names and codes unique	PASS	
Layers	Every layer: ACL, linetype (loaded), lineweight, plot flag, description	PASS	34/34 correct

Area	Test	Status	Evidence
Layers	No-plot layers exactly: ASG-CONSTRUCTION, ASG-NOPLOT, ASG-VIEWPORT	PASS	
Layers	Current layer is plottable geometry	PASS	ASG-OUTLINE-PRIMARY
Text styles	ASG-TEXT-ANNO: arial.ttf, height 0, annotative=True	PASS	
Text styles	ASG-TEXT-NOTE: arial.ttf, height 0, annotative=True	PASS	
Text styles	ASG-TEXT-SMALL: arial.ttf, height 0, annotative=True	PASS	
Text styles	ASG-TEXT-TITLE: arialbd.ttf, height 0, annotative=False	PASS	
Text styles	ASG-TEXT-HEADING: arialbd.ttf, height 0, annotative=False	PASS	
Text styles	ASG-TEXT-SHEET: arial.ttf, height 0, annotative=False	PASS	
Text styles	Annotative text behaviour at viewport scales	UNVERIFIED	Annotative flag written & re-read; scaling behaviour needs AutoCAD (README step 5)
Dimension styles	ASG-DIM-ANNO	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-FIXED	PASS	all 44 variables match; annotative=False
Dimension styles	ASG-DIM-DETAIL	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-ANGULAR	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-RADIUS	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-DIAMETER	PASS	all 44 variables match; annotative=True
Multileader styles	ASG-ML-ANNO	PASS	ASG-TEXT-NOTE 2.5 mm, arrow closed filled 2.5
Multileader styles	ASG-ML-NOTE	PASS	ASG-TEXT-NOTE 2.5 mm, arrow closed filled 2.5
Multileader styles	ASG-ML-DETAIL	PASS	ASG-TEXT-SMALL 1.8 mm, arrow_DOT 1.5
Multileader styles	ASG-ML-REFERENCE	PASS	ASG-TEXT-NOTE 2.5 mm, arrow_OPEN30 2.5
Multileader styles	Background mask	UNVERIFIED	Mask is an MText property, not stored in MLEADERSTYLE - documented per-note procedure
Multiline styles	ASG-MLINE-2	PASS	
Multiline styles	ASG-MLINE-3	PASS	
Annotation scales	Scale list = 1:1, 1:2, 1:5, 1:10, 1:20, 1:25, 1:50, 1:100	PASS	read back ['1:1', '1:2', '1:5', '1:10', '1:20', '1:25', '1:50', '1:100']; drawing units [1.0, 2.0, 5.0, 10.0, 20.0, 25.0, 50.0, 100.0]
Annotation scales	No duplicate scales	PASS	
Annotation scales	Scale list accepted by AutoCAD	UNVERIFIED	Raw AcDbScale objects; confirm in SCALELISTEDIT (ASG-QA checks it)
Table styles	7 ASG-TABLE-* styles	UNVERIFIED	ezdxf cannot write TABLESTYLE; created by ASG-SETUP in AutoCAD. In DXF now: none
Page setups	Named page setups ASG-	UNVERIFIED	ezdxf cannot write named PLOTSETTINGS; ASG-SETUP copies them from the layouts
Layouts	Layouts: A4-LANDSCAPE, A4-PORTRAIT, A3-LANDSCAPE, A3-PORTRAIT, A1-LANDSCAPE	PASS	['Model', 'A4-LANDSCAPE', 'A4-PORTRAIT', 'A3-LANDSCAPE', 'A3-PORTRAIT', 'A1-LANDSCAPE']
Layouts	A4-LANDSCAPE: 297x210 mm, DWG To PDF.pc3, layout area, 1:1, centred, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A4_(297.00_x_210.00_MM)
Title blocks	A4-LANDSCAPE: ASG-TB-A4 at scale 1 (not stretched), 15 attributes	PASS	
Sheets	A4-LANDSCAPE: border 20/10 mm margins inside 297x210	PASS	
Viewports	A4-LANDSCAPE: 1 viewport, locked, ASG-VIEWPORT (no plot), 1:10	PASS	
Viewports	A4-LANDSCAPE: viewport clear of title block	PASS	
Layouts	A4-PORTRAIT: 210x297 mm, DWG To PDF.pc3, layout area, 1:1, centred, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A4_(210.00_x_297.00_MM)
Title blocks	A4-PORTRAIT: ASG-TB-A4 at scale 1 (not stretched), 15 attributes	PASS	
Sheets	A4-PORTRAIT: border 20/10 mm margins inside 210x297	PASS	
Viewports	A4-PORTRAIT: 1 viewport, locked, ASG-VIEWPORT (no plot), 1:10	PASS	
Viewports	A4-PORTRAIT: viewport clear of title block	PASS	
Layouts	A3-LANDSCAPE: 420x297 mm, DWG To PDF.pc3, layout area, 1:1, centred, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A3_(420.00_x_297.00_MM)

Area	Test	Status	Evidence
Title blocks	A3-LANDSCAPE: ASG-TB-A3 at scale 1 (not stretched), 15 attributes	PASS	
Sheets	A3-LANDSCAPE: border 20/10 mm margins inside 420x297	PASS	
Viewports	A3-LANDSCAPE: 1 viewport, locked, ASG-VIEWPORT (no plot), 1:20	PASS	
Viewports	A3-LANDSCAPE: viewport clear of title block	PASS	
Layouts	A3-PORTRAIT: 297x420 mm, DWG To PDF.pc3, layout area, 1:1, centred, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A3_(297.00_x_420.00_MM)
Title blocks	A3-PORTRAIT: ASG-TB-A3 at scale 1 (not stretched), 15 attributes	PASS	
Sheets	A3-PORTRAIT: border 20/10 mm margins inside 297x420	PASS	
Viewports	A3-PORTRAIT: 1 viewport, locked, ASG-VIEWPORT (no plot), 1:20	PASS	
Viewports	A3-PORTRAIT: viewport clear of title block	PASS	
Layouts	A1-LANDSCAPE: 841x594 mm, DWG To PDF.pc3, layout area, 1:1, centred, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A1_(841.00_x_594.00_MM)
Title blocks	A1-LANDSCAPE: ASG-TB-A1 at scale 1 (not stretched), 15 attributes	PASS	
Sheets	A1-LANDSCAPE: border 20/10 mm margins inside 841x594	PASS	
Viewports	A1-LANDSCAPE: 1 viewport, locked, ASG-VIEWPORT (no plot), 1:50	PASS	
Viewports	A1-LANDSCAPE: viewport clear of title block	PASS	
Title blocks	ASG-TB-A4: 15 attribute definitions, 180 x 63.7 mm	PASS	
Title blocks	ASG-TB-A3: 15 attribute definitions, 180 x 81.9 mm	PASS	
Title blocks	ASG-TB-A1: 15 attribute definitions, 250 x 109.2 mm	PASS	
Model space	Model space empty	PASS	0 entities
References	No external references	PASS	
References	No image / PDF underlay dependencies	PASS	
Interoperability	No proxy entities	PASS	
Scope	Only title-block blocks (no product blocks / libraries)	PASS	['ASG-TB-A1', 'ASG-TB-A3', 'ASG-TB-A4']
Interoperability	Fonts referenced	PASS	['arial.ttf', 'arialbd.ttf'] - Windows core fonts; AutoCAD substitutes via FONTALT if missing
Automation	Re-running scale injection adds nothing (idempotent)	PASS	
Automation	Rebuild produces identical layer / style / layout sets	PASS	
Automation	Re-running layout creation creates no duplicate layouts	PASS	
Automation	ASG_Setup.lsp idempotency	UNVERIFIED	Uses get-or-create; needs AutoCAD
Drawing templates	ASG_A4-LANDSCAPE.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A4-PORTRAIT.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A3-LANDSCAPE.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A3-PORTRAIT.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A1-LANDSCAPE.dxf: opens, single layout, full standard, empty model	PASS	
Plot style	ASG_Monochrome.ctb: binary header (4-byte fields) + zlib body re-read	PASS	ezdxf writer bug (8-byte fields on 64-bit Linux) worked around and verified
Plot style	CTB: ACI 1-249,255 -> black; 250-254 -> grey; object lineweight	PASS	
Plot style	CTB loads in AutoCAD Plot Style Manager	UNVERIFIED	README step 1
QA drawing	Temporary QA drawing is a separate file	PASS	QA_TEST_DRAWING_temporary.dxf
Dimensions	Linear dims measure geometry (1200, 2400)	PASS	[1200.0, 2400.0]
Dimensions	Detail dim measures 50	PASS	
Dimensions	Angular dim measures 30.00 deg	PASS	30.0
Dimensions	Radius / diameter dims present and measure R40 / D80	PASS	R [40.0] D [80.0]
Dimensions	No overridden dimension text	PASS	
Dimensions	Rendered dim text precision: 1200 / 2400 (0 dp), 50.0 (detail 0.0), angles 0.00	PASS	['1200', '2400', '50.0', '30.00°', 'R40', 'Ø80']

Area	Test	Status	Evidence
Viewports	QA viewports at 1:20 and 1:5	PASS	[5.0, 20.0]
Lineweights	Lineweight matrix (2400 dpi render): 8 classes distinct, ordered, proportional	PARTIAL	nominal->rendered width ratio: 0.50:0.36, 0.40:0.34, 0.35:0.36, 0.30:0.35, 0.25:0.34, 0.18:0.35, 0.13:0.33, 0.09:0.35. Hierarchy confirmed; absolute plotted widths need AutoCAD PDF plot (renderer not mm-calibrated).
Plotting	No-plot layers (ASG-CONSTRUCTION, ASG-NOLOT) excluded from plot	UNVERIFIED	Plot flag is set and re-read (PASS above), but the ezdxf renderer ignores the layer plot flag - no-plot geometry appears in QA_render_A3. Confirm in AutoCAD plot preview.
Plotting	Viewport boundaries not plotted	UNVERIFIED	ASG-VIEWPORT plot flag off (PASS above); renderer never draws viewport frames, so not testable here
Plotting	PDF output via DWG To PDF.pc3 with CTB	UNVERIFIED	QA_render_A3.pdf produced by ezdxf/matplotlib, not by AutoCAD plot engine
Annotation scales	Annotative objects in viewports (text/dims/mleaders)	UNVERIFIED	ezdxf cannot create scale-context data; QA drawing uses fixed-scale overrides; test in AutoCAD
Deliverables	ASG_Layer_Register.xlsx written and re-opened	PASS	
Deliverables	ASG_CAD_Standards.pdf written	PASS	

QA render - temporary QA drawing (A3, viewports 1:20 and 1:5, lineweight matrix, text heights). Not part of the master template.

TEXT 2.5 @1:20

2400

1200

NO PLOT

R40

Ø80

50.0

30.00°

0.50

ASG-BORDER

0.40

ASG-SECTION-CUT

0.35

ASG-OUTLINE-PRIMARY

0.30

ASG-MILD-STEEL

0.25

ASG-ALUMINIUM-PROFILE

0.18

ASG-OUTLINE-SECONDARY

0.13

ASG-GASKET

0.09

ASG-HATCH

2.5 mm General dimensions

2.5 mm Technical notes

2.5 mm Material labels

1.8 mm Small detail notes

3.5 mm Detail titles

5.0 mm Drawing titles

7.0 mm Sheet headings

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.

AL SIRAJ GROUP | ALUMINIUM, GLASS & FACADE DIVISION | UNITED ARAB EMIRATES

PROJECT	LOCATION	
PROJECT NAME	LOCATION, UAE	
CLIENT	CONSULTANT	
CLIENT	CONSULTANT	

DRAWING TITLE

DRAWING TITLE

DRAWN BY	CHECKED BY	APPROVED BY
-	-	-
SCALE	DATE	DRAWING STATUS
AS SHOWN	DD.MM.YYYY	PRELIMINARY

DRAWING No.	REV	SHEET No.
AS-YYYY-NNN-DIS-TYP-001	R0	01 OF 01